
Pure Relative Structured Matrix Approximation with Square-Root Matvecs

Abstract

Let $\mathcal{L} \subseteq \mathbb{R}^{n \times m}$ be any known q -dimensional linear matrix family, and let an unknown A be accessible only through products Ax and $A^\top y$. A recent result obtains a nearly optimal approximation from $\mathcal{L}$ in about $\sqrt{q}$ queries, but incurs factor three and additive error proportional to $\|A\|_F$. Whether sublinear query complexity can give pure $(1 + \epsilon)$ relative error was left open.

We resolve this problem with a polynomial-time, nonadaptive algorithm. For a Frobenius-orthonormal basis $B_1, \dots, B_q$, form the basis-invariant partial trace $S_L = \sum_i B_i B_i^\top$. The algorithm queries the leading eigenspace of S_L exactly from the left, then fits the remaining directions from a Gaussian right sketch. If λ_j are the eigenvalues of S_L , its constant-success query complexity is

$$\min_s \left\{ s + C \left[\max\{1, \lambda_{s+1}\} \log^2(2q) \right] + \lambda_{s+1}/\epsilon \right\}.$$

The transposed profile is also available. Since $\lambda_{s+1} \leq q/(s+1)$, every family admits pure relative error with $\tilde{O}(\sqrt{q}/\epsilon)$ queries, independent of ambient dimensions and $\|A\|_F/\text{OPT}$. The key proof is a partial-trace covariance lemma for a Gaussian matrix-valued design. Exact high-leverage queries make its expected Hessian the identity, while the centered regression offset has variance only $O(\lambda_{s+1} \text{OPT}^2/k)$. A median construction amplifies success without estimating residual norms. Finally, a symmetric Wishart lower bound for fixed-sparsity approximation yields $\Omega(\sqrt{q}/\epsilon)$ adaptive two-sided queries when $q\epsilon$ is bounded below. Thus the universal rate is optimal up to logarithmic factors in this joint regime.

1 Introduction

Matrix-vector products are the basic access primitive for implicit Hessians, linear solution operators, kernel matrices, and matrix functions. When a matrix A is too large to form, one may still apply $x \mapsto Ax$ and, when an adjoint is available, $y \mapsto A^\top y$. This motivates a statistical question: how many such queries reveal the best structured approximation to A ?

We study a known linear matrix family $\mathcal{L} \subseteq \mathbb{R}^{n \times m}$ of dimension q . Given two-sided matvec access to an arbitrary A , the goal is to return $\hat{B} \in \mathcal{L}$ such that

$$\|A - \hat{B}\|_F \leq (1 + \epsilon) \min_{B \in \mathcal{L}} \|A - B\|_F. \quad (1)$$

The target is agnostic. No stochastic noise model or promise that $A \in \mathcal{L}$ is imposed.

Amsel et al. (2026a) initiated a general theory for this problem. For a finite family $\mathcal{F}$, they give a nearly quadratic improvement from $O(\log |\mathcal{F}|)$ one-sided queries to $\tilde{O}(\sqrt{\log |\mathcal{F}|})$ two-sided queries. Covering a linear family gives roughly $\sqrt{q}$ queries, but the guarantee is

$$(3 + \epsilon) \text{OPT} + \alpha \|A\|_F. \quad (2)$$

They explicitly ask whether every q -dimensional linear family admits pure $(1 + \epsilon)$ error with $\tilde{O}(\sqrt{q}/\text{poly}(\epsilon))$ queries, and whether this can be achieved in polynomial time.

The obstruction is not identifiability. In the realizable case, generic queries often recover a linear family exactly (Halikias and Townsend, 2024). The difficulty is an arbitrary orthogonal residual. A direct Gaussian least-squares fit has an unbiased normal equation, but a few matrix contractions need not embed all q parameter directions. Requiring a conventional subspace embedding returns to order q queries.

Our solution separates the directions that actually obstruct concentration. Let $B_1, \dots, B_q$ be a Frobenius-orthonormal basis of $\mathcal{L}$ and define

$$S_L = \sum_{i=1}^q B_i B_i^\top, \quad S_R = \sum_{i=1}^q B_i^\top B_i. \quad (3)$$

These partial traces are basis invariant, positive semidefinite, and have trace q . We query a leading eigenspace of one partial trace exactly. On its orthogonal complement, Gaussian matvecs have both a well-conditioned Hessian and a small regression offset. The first fact costs the remaining spectral leverage times logarithmic factors. The second costs the same leverage divided by ϵ . Optimizing where to split gives square-root complexity.

Contributions. Our results are entirely theoretical.

- We give a nonadaptive, polynomial-time algorithm satisfying (1) for every rectangular linear family. Its query complexity is the better of two basis-invariant spectral profiles, one for each query orientation.
- The profile implies a universal $\tilde{O}(\sqrt{q/\epsilon})$ upper bound. It has no additive error and no dependence on the ambient dimensions or the ratio $\|A\|_F/\text{OPT}$.
- We prove a Gaussian covariance lemma controlled by a partial trace. The proof combines a dimension-free Schatten moment estimate, an exact fourth-moment calculation, and matrix concentration for unbounded summands.
- We convert the adaptive symmetric Wishart lower bound of Amsel et al. (2026b) into $\Omega(\sqrt{q/\epsilon})$ for general q -dimensional linear families when $q\epsilon = \Omega(1)$. This matches our upper bound up to logarithms. A separate GOE construction shows that $\Omega(1/\sqrt{\epsilon})$ queries can be necessary even for a one-dimensional family.

Why two sides matter. The partial trace identifies output directions shared by many basis matrices. A left query reveals one such direction exactly, no matter how many coefficients use it. Gaussian right queries then handle the diffuse remainder. Strong one-sided barriers are known for general matrix-free tasks (Halikias et al., 2026). Our estimator makes the complementary roles of the two orientations explicit.

Table 1 separates the guarantee and query dependence of the closest general result. The earlier linear-family bound is obtained by covering a bounded part of $\mathcal{L}$, then running a finite-candidate procedure. Its runtime scales with the cover size. Our algorithm instead solves linear least squares in the original q coefficients.

2 Why a Spectral Split Is Necessary

It is tempting to fit $\mathcal{L}$ using only a right Gaussian sketch, or only a left sketch, then choose whichever ori-

Table 1: General q -dimensional linear families. Logarithmic factors other than $\log(1/\alpha)$ are suppressed.

Result	Queries	Guarantee
Amsel et al.	$\frac{\sqrt{q \log(1/\alpha)}}{\epsilon^2}$	$(3 + \epsilon)\text{OPT} + \alpha\ A\ _F$
This paper	$\sqrt{q/\epsilon}$	$(1 + \epsilon)\text{OPT}$

entation has smaller aggregate variance. This cannot prove a square-root theorem. For an orthonormal basis, define the two offset leverages of a residual $R \perp \mathcal{L}$ by

$$V_R(R) = \sum_i \|R^\top B_i\|_F^2, \quad V_L(R) = \sum_i \|RB_i^\top\|_F^2. \quad (4)$$

A one-sided Gaussian normal equation has total offset variance proportional to the corresponding quantity. One might hope that $\min\{V_R, V_L\} = O(\sqrt{q}\|R\|_F^2)$, or that their product is at most $q\|R\|_F^4$. Both statements are false.

Proposition 1 (Deleted-tangent obstruction). *For every $a, b \geq 1$, there is a linear family of dimension $q = a + b$ and a unit residual $R \perp \mathcal{L}$ such that*

$$V_R(R) = b, \quad V_L(R) = a.$$

When a and b are comparable, both global orientations have order- q offset variance. Nevertheless, the spectral-split estimator removes all contamination with one exact query.

Proof. Let $u, e_1, \dots, e_a$ and $v, f_1, \dots, f_b$ be orthonormal systems. Take $R = uv^\top$ and let $\mathcal{L}$ have orthonormal basis

$$\{uf_j^\top : 1 \leq j \leq b\} \cup \{e_i v^\top : 1 \leq i \leq a\}. \quad (5)$$

The deleted matrix $uv^\top$ is orthogonal to every basis element. For the first arm, $\|R^\top(uf_j^\top)\|_F = 1$, while the corresponding terms from the second arm vanish. This gives $V_R = b$. The transposed calculation gives $V_L = a$.

The left partial trace is

$$S_L = buu^\top + \sum_{i=1}^a e_i e_i^\top.$$

Querying u exactly leaves partial-trace norm one. More strongly, the remaining residual is $(I - uu^\top)R = 0$, so the random-stage offset vanishes. $\square$

This example isolates the role of the deterministic term in (6). It is not included to obtain a conventional subspace embedding. It makes the high-curvature and

high-offset directions exact, which leaves a centered problem whose variance is governed by the residual partial trace. A fixed global choice of orientation misses this local structure.

3 Problem and Algorithm

Let $\mathcal{L} \subseteq \mathbb{R}^{n \times m}$ have Frobenius-orthonormal basis $B_1, \dots, B_q$. Write

$$B_\star = P_{\mathcal{L}}A, \quad R = A - B_\star, \quad \text{OPT} = \|R\|_F.$$

Then $\langle R, B \rangle_F = 0$ for every $B \in \mathcal{L}$. For the left partial trace S_L , let $\lambda_1^L \geq \lambda_2^L \geq \dots \geq 0$ be its eigenvalues, padded by zeros. Fix $s \in \{0, \dots, n\}$, let P project onto its first s eigenvectors, set $Q = I - P$, and abbreviate $\lambda = \lambda_{s+1}^L = \|QS_LQ\|_{\text{op}}$.

If $\lambda = 0$, query $A^\top$ on an orthonormal basis of $\text{range}(P)$ and minimize $\|P(A - B)\|_F^2$ over $B \in \mathcal{L}$. This recovers $B_\star$ because $QB = 0$ for every $B \in \mathcal{L}$.

Suppose $\lambda > 0$. In addition to the same s left queries, draw $G = [g_1, \dots, g_k] \in \mathbb{R}^{m \times k}$ with independent standard Gaussian entries and issue the k right queries Ag_ℓ . Return the minimum-norm minimizer

$$\hat{B} \in \arg \min_{B \in \mathcal{L}} \left\{ \|P(A - B)\|_F^2 + \frac{1}{k} \|Q(A - B)G\|_F^2 \right\}. \quad (6)$$

The objective is observable. The left responses give PA , and QAG is obtained from $AG - PAG$. Equation (6) is an ordinary q -variable least-squares problem.

Define the one-orientation profile

$$\mathbf{Q}_L(\epsilon) = \min_{0 \leq s \leq n} \left\{ s + C \left[\max\{1, \lambda_{s+1}^L\} \log^2(2q) + \frac{\lambda_{s+1}^L}{\epsilon} \right] \right\}, \quad (7)$$

where the bracketed term is zero if $\lambda_{s+1}^L = 0$. Define $\mathbf{Q}_R(\epsilon)$ analogously from S_R . The right profile swaps the roles of A and $A^\top$.

Theorem 2 (Pure relative profile). *For every $A \in \mathbb{R}^{n \times m}$, every q -dimensional linear family $\mathcal{L}$, and every $\epsilon \in (0, 1)$, the better orientation of (6) returns $\hat{B} \in \mathcal{L}$ satisfying (1) with probability at least 0.9 using at most*

$$\min\{\mathbf{Q}_L(\epsilon), \mathbf{Q}_R(\epsilon)\}$$

matvec queries. The query vectors are nonadaptive and the estimator runs in time polynomial in the explicit input size of the basis.

For failure probability δ , multiply only the bracketed random-query term in (7) by $C \log(1/\delta)$, and likewise for the right profile.

The leading eigenspace is not merely convenient. It is the best rank- s choice for this estimator. For an arbitrary rank- s orthogonal projector P , define its residual leverage

$$\Lambda_L(P) = \|(I - P)S_L(I - P)\|_{\text{op}}. \quad (8)$$

The same proof as Theorem 2 gives query count

$$s + C \left[\max\{1, \Lambda_L(P)\} \log^2(2q) + \frac{\Lambda_L(P)}{\epsilon} \right]. \quad (9)$$

Proposition 3 (Optimal exact-query subspace). *Among all rank- s orthogonal projectors P ,*

$$\min_{\text{rank}(P)=s} \Lambda_L(P) = \lambda_{s+1}^L.$$

Thus the leading eigenspace minimizes both random-query terms in (9).

Proof. For $Q = I - P$, $\Lambda_L(P)$ is the largest Rayleigh quotient of S_L on $\text{range}(Q)$. The claim is the Courant–Fischer characterization of λ_{s+1}^L . $\square$

Corollary 4 (Universal square-root bound). *Let $L_\epsilon = \log^2(2q) + 1/\epsilon$. Constant-success pure relative error requires at most*

$$O\left(\sqrt{qL_\epsilon} + \log^2(2q)\right) = \tilde{O}\left(\sqrt{q/\epsilon}\right) \quad (10)$$

two-sided matvec queries.

Proof. The partial-trace eigenvalues sum to q , so $\lambda_{s+1}^L \leq q/(s+1)$. Also $\max\{1, \lambda\} \leq 1 + \lambda$. Insert these inequalities into (7) and choose $s = \min\{n, \lfloor \sqrt{qL_\epsilon} \rfloor\}$. If this choice reaches n , the exact queries already cover the row support of $\mathcal{L}$. Otherwise the two s -dependent terms are both $O(\sqrt{qL_\epsilon})$. $\square$

Examples. The profile is more informative than the universal bound. For a full $a \times b$ block family, $q = ab$, and exact recovery uses $\min\{a, b\}$ queries. A balanced block gives the familiar $\sqrt{q}$ barrier. A row-star family is exact after one left query. For the diagonal family, $S_L = I$ and the randomized stage uses $O(\log^2 q + 1/\epsilon)$ queries, matching the dimension-free behavior of specialized diagonal estimators up to logarithms (Amsel et al., 2026b; Dharangutte and Musco, 2023). For a coordinate subspace, the partial traces are the row and column degree matrices. The algorithm exactly removes high-degree rows or columns before sketching the rest. This is a spectral analogue of sparse-pattern peeling, although graph degeneracy can give a sharper profile for particular masks (Musco and Ramesh, 2026).

4 Proof of the Upper Bound

We give the main argument and defer moment details and amplification to the supplement. Set $C_i = QB_i$ and define the $q \times q$ Gram matrix

$$K_{ij} = \langle C_i, C_j \rangle_F.$$

Since the original basis is orthonormal, $0 \preceq K \preceq I$. Moreover,

$$\sum_{i=1}^q C_i C_i^\top = Q S_L Q \preceq \lambda I. \quad (11)$$

For $g \sim N(0, I_m)$, let $D(g) \in \mathbb{R}^{n \times q}$ have columns $C_i g$.

Lemma 5 (Partial-trace Gaussian covariance). *Suppose $K \preceq I$ and $\sum_i C_i C_i^\top \preceq \tau I$ for $\tau \geq 1$. There is a universal constant C such that, if $g_1, \dots, g_k$ are independent Gaussians and*

$$k \geq C\tau \log^2(2q),$$

then, with probability at least 0.99,

$$\left\| \frac{1}{k} \sum_{\ell=1}^k D(g_\ell)^\top D(g_\ell) - K \right\|_{\text{op}} \leq \frac{1}{2}. \quad (12)$$

Proof. The ambient dimensions disappear through Schatten moments. Write $D(g) = \sum_a g_a A_a$, where $A_a = [C_1 e_a, \dots, C_q e_a]$. Direct calculation gives

$$\sum_a A_a A_a^\top = \sum_i C_i C_i^\top \preceq \tau I, \quad \sum_a A_a^\top A_a = K \preceq I. \quad (13)$$

Both matrices have trace $\text{tr } K \leq q$. Hence, for every integer $p \geq 1$,

$$\text{tr} \left(\sum_a A_a A_a^\top \right)^p \leq \tau^{p-1} q, \quad \text{tr } K^p \leq q. \quad (14)$$

Rectangular noncommutative Khintchine (Buchholz, 2001), followed by $\|\cdot\|_{\text{op}} \leq \|\cdot\|_{S_{2p}}$, now gives

$$(\mathbb{E} \|D(g)\|_{\text{op}}^{2p})^{1/(2p)} \leq C\sqrt{p\tau}, \quad p \geq \log(2q). \quad (15)$$

The factors $q^{1/(2p)}$ from (14) are constant. This is the step that removes n and m .

Set $Z = D(g)^\top D(g)$, so $\mathbb{E} Z = K$. For a unit $\theta \in \mathbb{R}^q$, write $C_\theta = \sum_i \theta_i C_i$. Then

$$\theta^\top \mathbb{E} Z^2 \theta = \sum_i \mathbb{E} \langle C_i g, C_\theta g \rangle^2. \quad (16)$$

For a real Gaussian, the identity

$$\mathbb{E}(g^\top M g)^2 = (\text{tr } M)^2 + \text{tr}(M^2) + \text{tr}(M M^\top)$$

and Cauchy–Schwarz yield

$$\begin{aligned} \theta^\top \mathbb{E} Z^2 \theta &\leq \theta^\top K^2 \theta + 2 \sum_i \|C_i^\top C_\theta\|_F^2 \\ &= \theta^\top K^2 \theta + 2 \text{tr} \left(C_\theta^\top \sum_i C_i C_i^\top C_\theta \right) \\ &\leq \theta^\top (K^2 + 2\tau K) \theta. \end{aligned} \quad (17)$$

Consequently,

$$\mathbb{E}(Z - K)^2 \preceq 2\tau K \preceq 2\tau I. \quad (18)$$

For independent copies $X_\ell = Z_\ell - K$, the variance parameter of $\sum_\ell X_\ell$ is at most $2k\tau$. Equation (15) and $\|X_\ell\|_{\text{op}} \leq \|D(g_\ell)\|_{\text{op}}^2 + 1$ also imply

$$\left(\mathbb{E} \max_{\ell \leq k} \|X_\ell\|_{\text{op}}^2 \right)^{1/2} \leq C\tau \log(2qk). \quad (19)$$

Indeed, take the L_{2p} norm with $p = \lceil \log(2qk) \rceil$ and use the union bound in moment form.

The expected-norm inequality of Tropp (2015) applied to the centered $q \times q$ matrices X_ℓ gives

$$\begin{aligned} &\left(\mathbb{E} \left\| \frac{1}{k} \sum_{\ell=1}^k X_\ell \right\|_{\text{op}}^2 \right)^{1/2} \\ &\leq C \sqrt{\frac{\tau \log(2q)}{k}} + \frac{C\tau \log(2q) \log(2qk)}{k}. \end{aligned} \quad (20)$$

For $k = C_0 \tau \log^2(2q)$ and sufficiently large C_0 , the right side is at most $1/20$. It decreases for larger k . Markov's inequality proves (12). Supplement A gives the arbitrary-accuracy version. $\square$

The second ingredient controls only the centered offset, which is why the accuracy dependence is $1/\epsilon$ rather than $1/\epsilon^2$.

Lemma 6 (Offset variance). *For any fixed $M \in \mathbb{R}^{n \times m}$, define*

$$\xi_i = \frac{1}{k} \sum_{\ell=1}^k \{ \langle M g_\ell, C_i g_\ell \rangle - \langle M, C_i \rangle_F \}.$$

Under (11),

$$\mathbb{E} \|\xi\|_2^2 \leq \frac{2\lambda}{k} \|M\|_F^2. \quad (21)$$

Proof. The real Gaussian quadratic-form identity implies $\text{Var}(\langle M g, C_i g \rangle) \leq 2\|M^\top C_i\|_F^2$. Summing over i gives

$$\sum_i \|M^\top C_i\|_F^2 = \text{tr} \left(M^\top \sum_i C_i C_i^\top M \right) \leq \lambda \|M\|_F^2.$$

Independence across the k columns completes the proof. $\square$

Proof of Theorem 2. In orthonormal coordinates of $\mathcal{L}$, the Hessian of (6) is

$$H = H_P + \frac{1}{k} \sum_{\ell=1}^k D(g_\ell)^\top D(g_\ell),$$

$$(H_P)_{ij} = \langle PB_i, PB_j \rangle_F.$$

Since $H_P + K = I$, Lemma 5 with $\tau = \max\{1, \lambda\}$ gives $H \succeq I/2$.

At $B_\star$, exact orthogonality $\langle R, B_i \rangle_F = 0$ makes the gradient equal to the centered vector in Lemma 6, with $M = QR$. If d is the coefficient vector of $\hat{B} - B_\star$, the normal equation is $Hd = \xi$. On the Hessian event,

$$\|\hat{B} - B_\star\|_F^2 = \|d\|_2^2 \leq 4\|\xi\|_2^2.$$

Lemma 6 and Markov's inequality show that $k \geq C\lambda/\epsilon$ makes this at most $2\epsilon\text{OPT}^2$ with constant probability. Finally, orthogonality gives the exact identity

$$\|A - \hat{B}\|_F^2 = \text{OPT}^2 + \|\hat{B} - B_\star\|_F^2. \quad (22)$$

Since $\sqrt{1+2\epsilon} \leq 1 + \epsilon$, the constant-success result follows. $\square$

We use the metric-median device of Amsel et al. (2026b). Naive confidence amplification would estimate each candidate's residual norm. A Frobenius sketch accurate enough to distinguish $(1 + \epsilon)$ error costs $O(1/\epsilon^2)$ extra queries, which would destroy the rate. The following metric argument avoids validation entirely.

Lemma 7 (Validation-free amplification). *Run the Gaussian stage independently an odd number r of times, reusing the same exact responses. For candidate i , let ρ_i be its median Frobenius distance to the r candidates, including itself. Return a candidate minimizing ρ_i . If each candidate satisfies*

$$\|\hat{B}_i - B_\star\|_F^2 \leq \gamma\text{OPT}^2 \quad (23)$$

with probability at least $19/20$, then $r = C \log(1/\delta)$ gives

$$\|\hat{B} - B_\star\|_F^2 \leq 9\gamma\text{OPT}^2$$

with probability at least $1 - \delta$.

Proof. A Chernoff bound gives a strict majority of candidates satisfying (23). Condition on this event. Every good candidate has median radius at most $2\sqrt{\gamma}\text{OPT}$. The selected candidate has no larger radius. Its median ball and the strict majority of good candidates must intersect. The triangle inequality through a candidate in the intersection gives distance at most $3\sqrt{\gamma}\text{OPT}$ from $B_\star$. $\square$

Set $\gamma = 2\epsilon/9$ and apply (22). This proves the confidence statement in Theorem 2. Pairwise distances are computed from the known coefficient vectors, so amplification uses no new kind of oracle query.

5 Lower Bounds and Optimality

The square-root dependence is already necessary for exact recovery. Take the family of all $r \times r$ matrices, so $q = r^2$. Under a continuous Gaussian prior, each adaptive left or right query imposes at most r linear constraints. Fewer than r queries leave a non-atomic posterior on a positive-dimensional affine space. Since pure relative error must be exact when $\text{OPT} = 0$, Yao's principle gives $\Omega(\sqrt{q})$ queries.

Accuracy alone also forces a square-root rate, even for one parameter. This endpoint is useful because the fixed-sparsity construction below requires $q\epsilon$ to be bounded below.

Proposition 8 (One-parameter accuracy lower bound). *For every sufficiently small ϵ , there are an ambient dimension d and a one-dimensional family $\mathcal{L}$ such that every possibly randomized and adaptive two-sided algorithm needs $\Omega(1/\sqrt{\epsilon})$ queries for a constant-probability $(1 + \epsilon)$ approximation.*

Proof. Let $\mathcal{L} = \text{span}\{I_d/\sqrt{d}\}$ and draw A from the Gaussian orthogonal ensemble. Thus the law is invariant under $A \mapsto U^\top AU$, its diagonal entries have variance two, and its upper off-diagonal entries have variance one. Since $A = A^\top$, transpose queries give no extra access.

Fix a deterministic adaptive algorithm. New queries may be orthogonalized against earlier ones because the response on their span is already known. After t orthonormal queries, rotate them to the first t coordinates. The transcript determines the first t rows and columns, while the bottom-right $(d - t) \times (d - t)$ block remains an independent GOE matrix. This remains true for adaptive queries by sequential conditioning and rotational invariance.

The optimal coefficient is $\beta_\star = \langle A, I_d/\sqrt{d} \rangle_F = \text{tr}(A)/\sqrt{d}$. Conditional on the transcript, its unknown part is Gaussian with variance $2(d - t)/d$. If $t \leq d/2$, this variance is at least one. Also $\mathbb{E}\|A\|_F^2 = d(d + 1)$. With probability at least 0.9, $\text{OPT}^2 \leq \|A\|_F^2 \leq 10d(d + 1)$. Any successful output must satisfy

$$(\hat{\beta} - \beta_\star)^2 \leq ((1 + \epsilon)^2 - 1)\text{OPT}^2 \leq 3\epsilon\text{OPT}^2. \quad (24)$$

Take $d = \lfloor c/\sqrt{\epsilon} \rfloor$ for a sufficiently small constant c . On the norm event, the right side of (24) is a small universal constant. Gaussian anti-concentration bounds the conditional success probability away from

one for every transcript-measurable estimate. Thus $t > d/2 = \Omega(1/\sqrt{\epsilon})$ is necessary. Yao's principle extends the conclusion to randomized algorithms. Supplement C gives explicit constants and the conditioning induction. $\square$

The dependence on ϵ can be coupled to q . The following result is a reparameterization of the adaptive fixed-sparsity lower bound of Amsel et al. (2026b).

Theorem 9 (Joint lower bound). *There are universal constants $c, c_0, c_1 > 0$ such that, for $\epsilon < c_0$ and $q\epsilon \geq c_1$, there is a q -dimensional linear matrix family for which every possibly randomized and adaptive two-sided algorithm using fewer than*

$$c\sqrt{q/\epsilon}$$

queries fails to return a $(1 + \epsilon)$ approximation with constant probability.

Proof. Choose a sparsity level u . The cited theorem constructs a symmetric $d \times d$ Wishart hard instance with $d = \Theta(u/\epsilon)$ for any fixed pattern having between constant multiples of u entries in every row and column. It requires $\Omega(u/\epsilon)$ adaptive forward queries. Symmetry makes a transpose query identical to a forward query.

Choose a cyclic u -regular pattern S . Its coordinate family

$$\mathcal{L}_0 = \{B : B = S \circ B\}$$

has dimension

$$q_0 = du = \Theta(u^2/\epsilon). \quad (25)$$

The source lower bound becomes

$$\Omega(u/\epsilon) = \Omega(\sqrt{q_0/\epsilon}). \quad (26)$$

For $q\epsilon$ above a universal constant, choose $u = \lfloor c_2\sqrt{q\epsilon} \rfloor$ with c_2 small enough. The constants in $d = \Theta(u/\epsilon)$ can be chosen so that $c_3q \leq q_0 \leq q$. Integer rounding changes only universal constants.

It remains to reach dimension exactly q . Set $p = q - q_0$, let $\mathcal{D}_p$ be the p -dimensional diagonal family on a disjoint block, and define

$$\mathcal{L} = \{B_0 \oplus D : B_0 \in \mathcal{L}_0, D \in \mathcal{D}_p\}.$$

Embed a hard matrix as $A' = A_0 \oplus 0$. A query at (x, z) returns $(A_0x, 0)$, so it is simulated by one query to A_0 . The same holds for a transpose query. If an output $\hat{B}_0 \oplus \hat{D}$ succeeds, then

$$\begin{aligned} \|A_0 - \hat{B}_0\|_F^2 &\leq \|A_0 - \hat{B}_0\|_F^2 + \|\hat{D}\|_F^2 \\ &\leq (1 + \epsilon)^2 \|A_0 - S \circ A_0\|_F^2. \end{aligned}$$

Restriction to the first block would therefore solve the source problem. Equation (26) and $q_0 \geq c_3q$ prove the claim. The supplement records the source theorem and rounding explicitly. $\square$

Theorems 2 and 9 match up to logarithmic factors whenever $q\epsilon = \Omega(1)$. We do not claim a matching lower bound in the edge regime $q\epsilon = o(1)$. The exact-recovery and GOE constructions still give separate square-root lower bounds in q and $1/\epsilon$, while the upper theorem remains valid throughout.

6 Related Work and Discussion

Fixed-sparsity approximation is a coordinate-subspace instance of our problem. Gaussian rowwise regression gives $O(s/\epsilon)$ queries for maximum row degree s , with a matching adaptive lower bound (Amsel et al., 2026b). More recent work characterizes exact sparse recovery by graph degeneracy and gives a nearly matching approximation algorithm (Musco and Ramesh, 2026). These methods exploit disjoint coordinate support. Our partial-trace profile applies to arbitrary dense and overlapping basis matrices.

The split between deterministic high leverage and randomized low leverage has an analogue in row-sampled least squares (Larsen and Kolda, 2022, 2026). In that setting, leverage belongs to scalar rows of one design and a sample reveals one response row. Here one matvec reveals a tensor contraction shared by all q parameters. The partial trace controls the resulting matrix-valued design. Classical active regression needs $\Theta(q/\epsilon)$ scalar labels (Chen and Price, 2019). Sharp sketch-and-solve theory likewise has excess proportional to design rank divided by sketch size (Epperly and Webber, 2026). The two-sided tensor structure is what permits a square-root improvement.

Our logarithmic factors come from controlling unbounded Gaussian covariance summands through non-commutative Khintchine and a general expected-norm inequality (Buchholz, 2001; Tropp, 2015). Removing them may require a sharper lower-tail argument tailored to the regularized Hessian. Another open direction is the exact minimax profile outside the regime of Theorem 9. The present result resolves the pure-relative existence and polynomial-time questions while leaving these finer issues.

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Checklist

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. **Yes.** Sections 3 and 4 specify the oracle, estimator, spectral profile, and proof ingredients.
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. **Yes.** Theorem 2 and Corollary 4 give query complexity, and Theorem 2 states polynomial runtime in the explicit basis representation.
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. **Not Applicable.** The contribution is theoretical and has no empirical claim.
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. **Yes.** Each theorem states its family, oracle, accuracy, probability, and adaptivity conditions.
 - (b) Complete proofs of all theoretical results. **Yes.** Proof cores are in the main paper and complete details are in the supplement.
 - (c) Clear explanations of any assumptions. **Yes.** Sections 1–3 explain two-sided access, the agnostic residual, and the role of the spectral split.
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). **Not Applicable.**
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). **Not Applicable.**
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). **Not Applicable.**
 - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). **Not Applicable.**
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
 - (a) Citations of the creator If your work uses existing assets. **Not Applicable.**
 - (b) The license information of the assets, if applicable. **Not Applicable.**
 - (c) New assets either in the supplemental material or as a URL, if applicable. **Not Applicable.**
 - (d) Information about consent from data providers/curators. **Not Applicable.**
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. **Not Applicable.**
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
 - (a) The full text of instructions given to participants and screenshots. **Not Applicable.**
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. **Not Applicable.**
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. **Not Applicable.**